Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the validation accuracy by dataset.

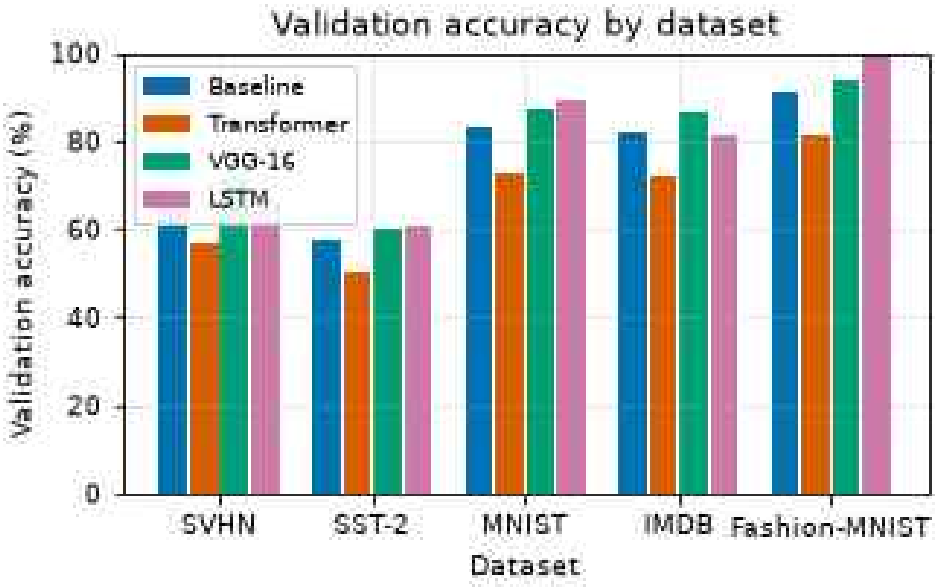


Figure 1: Validation accuracy by dataset for the models in this experiment.

On SST-2, LSTM is 13.4 points better than Baseline (Figure 1). Baseline achieves over 65% validation accuracy on SVHN. On MNIST, Baseline is 10.7 points better than Transformer.

Figure 2 shows the training loss.

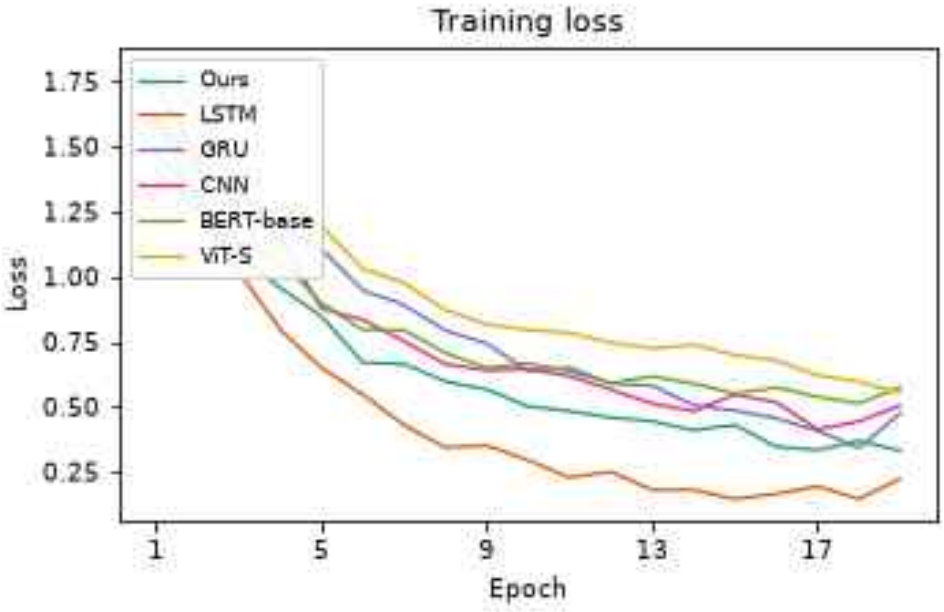


Figure 2: Training loss for the models in this experiment.

LSTM has a loss of 1.556 at epoch 5. At epoch 11, the loss of BERT-base is 0.272.

Figure 3 shows the validation accuracy by dataset.

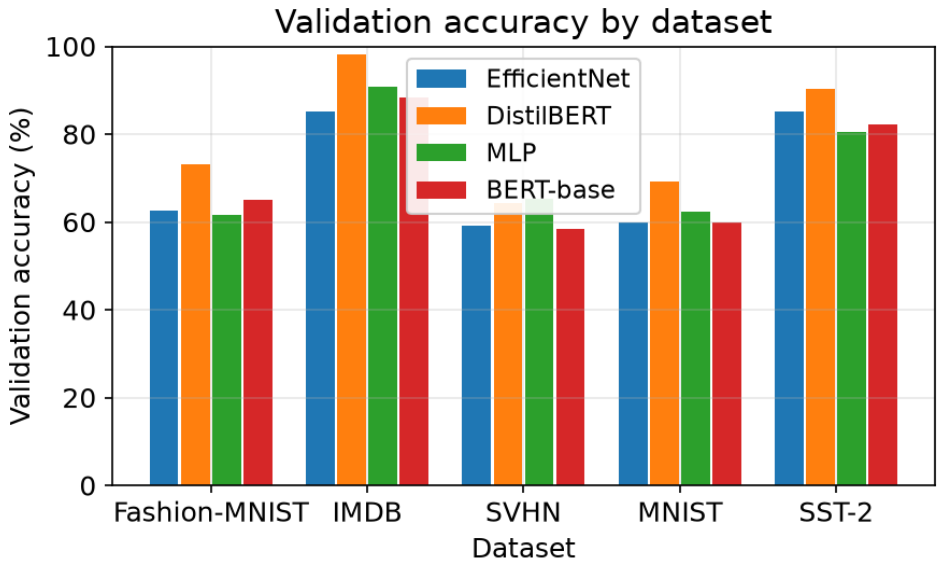


Figure 3: Validation accuracy by dataset for the models in this experiment.

On SVHN, EfficientNet obtains a validation accuracy of about 60% (Figure 3). On IMDB, EfficientNet obtains a validation accuracy of 88.4%. DistilBERT outperforms EfficientNet by 25.3 percentage points on SVHN.

Figure 4 shows the accuracy during training.

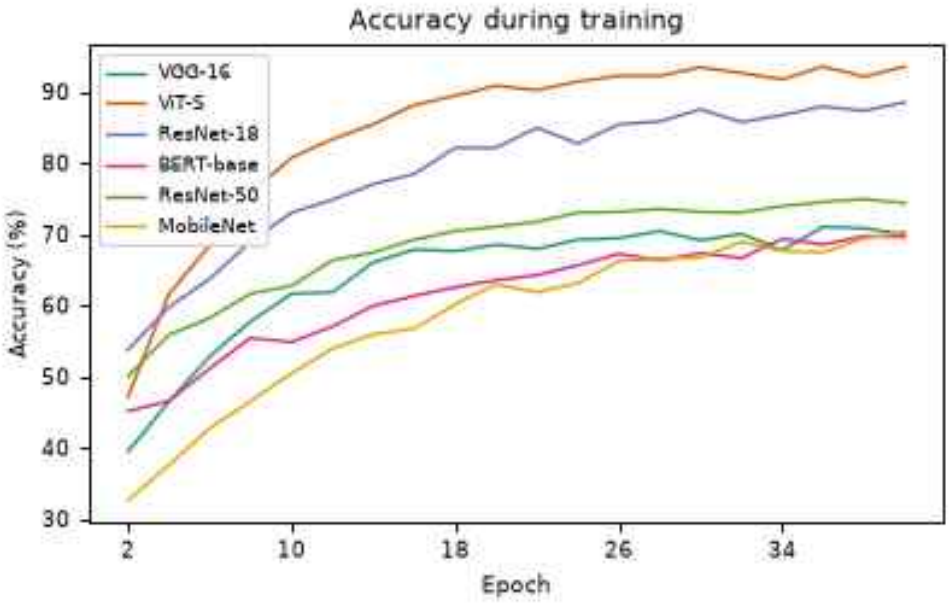


Figure 4: Accuracy during training for the models in this experiment.

From epoch 6 to epoch 16, the accuracy of BERT-base improves by a relative about 20%. ResNet-50 reaches 93.5% accuracy at epoch 40. From epoch 18 to epoch 30, the accuracy of ViT-S improves by a relative 19.5%.

3 Discussion

Overall, the results are in line with what we expected from the literature.